Problem 28

A steel ball is dropped from a height of $h = 1$ m onto a steel plate and bounces back. Each time it bounces, its velocity is reduced by 10%. The same reduction happens at each subsequent bounce. How long is the third fall time of the ball?

Solution

The time it takes for the ball to fall from a height h is determined using the free-fall equation*:

$$t = \sqrt{\frac{2h}{g}}.$$

For the initial drop from $h = 1$ m:

$$t_1 = \sqrt{\frac{2 \times 1}{9.81}} \approx \sqrt{0.204} \approx 0.45 \text{ s}.$$

After each bounce, the velocity is reduced by 10%, meaning the new velocity is 0.9 times the previous velocity. Since the maximum height after each bounce is proportional to the square of the velocity, the height after the first bounce is:

$$h_2 = (0.9)^2 h_1 = 0.81 \times 1 = 0.81 \text{ m}.$$

The time for the second fall:

$$t_2 = \sqrt{\frac{2 \times 0.81}{9.81}} \approx \sqrt{0.165} \approx 0.41 \text{ s}.$$

Similarly, the height for the third fall is:

$$h_3 = (0.9)^2 h_2 = 0.81 \times 0.81 = 0.6561 \text{ m}.$$

The time for the third fall:

$$t_3 = \sqrt{\frac{2 \times 0.6561}{9.81}} \approx \sqrt{0.134} \approx 0.37 \text{ s}.$$

Thus, the third fall time of the ball is approximately:

$$t_3 \approx 0.37 \text{ s}.$$